

# **DISASTER RELIEF RESOURCE MANAGEMENT SYSTEM**

## **Software Requirements Specification**

**Version 1.0**

**Group Id:** **write your group id here**

**Supervisor Name:** **write your supervisor name here**

### Revision History

| Date                            | Version | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Author              |
|---------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Write your submission date here | 1.0     | The Disaster Relief Resource Management System (DRRMS) is a web application revolutionizing humanitarian aid. It centralizes disaster response by tracking affected families, donations, and relief distribution end-to-end. DRRMS manages camp inventories, alerts administrators to urgent needs, and coordinates Camp Managers, Field Staff, and Volunteers. Through robust reporting, it enhances efficiency, transparency, and decision-making, ensuring critical aid is delivered swiftly and effectively to those most in need. | Writer your ID here |

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**Scope of Project:**

The Disaster Relief Resource Management System project encompasses the development of a comprehensive web-based application designed to streamline operations for humanitarian aid and disaster response. This system will include robust User Account Management capabilities, allowing for the registration, viewing, editing, and searching of profiles for Camp Managers, Field Staff, Volunteers, and Donors. Central to the application is the Donation & Relief Management Module, facilitating the creation of relief appeals, the logging of pledged contributions, and the lifecycle tracking of donations—from "Pledged" to "In-Transit" to "Distributed"—while also offering robust search and filter functionalities. For efficient aid distribution, a Camp Resource Management Module will enable the tracking of relief inventory (food, medical supplies, tents), automatically decreasing stock levels as resources are allocated to affected families, triggering low-stock alerts with urgent need notifications, and permitting manual stock adjustments by Camp Managers. Volunteer & Field Staff Coordination is also a key component, supporting the registration of volunteers, assignment of on-ground tasks, tracking of task completion, and monitoring of individual performance and availability. The application further integrates a Transparency & Auditing Module for automated donation receipt generation, recording of aid utilization, and tracking of resource distribution to ensure accountability. Critical for strategic decision-making, the Reports & Analytics Module will provide a dashboard and customizable reports on total donations vs. requirements, most urgently needed items, camp occupancy levels, and distribution efficiency, with export capabilities. Finally, secure User Authentication and Authorization will ensure role-based access control throughout the system.

This initial phase of the project specifically excludes features such as a direct payment gateway integration for real-time fund transfer, complex GIS routing for logistics trucks, drone-based damage assessment integration, or a dedicated mobile application (the web app will be mobile-responsive). Furthermore, biometric verification for victims and cross-border international aid compliance modules are also outside the current scope.

Key deliverables for this project will include the fully functional Disaster Relief web application, its complete database schema and initial data, detailed deployment instructions, a comprehensive user manual for both administrators and camp managers, and the full source code repository. The project operates under the assumptions of intermittent to reliable internet connectivity for camp managers, the availability of necessary server infrastructure (XAMPP/WAMP for local or Cloud hosting), and the provision of clear data regarding camp locations. Development is further constrained by a predefined timeline and the utilization of the PHP/MySQL technology stack.

### **Functional and Non-Functional Requirements:**

#### **FUNCTIONAL REQUIREMENTS: -**

- **Customer Management**

- **User Registration:** The system shall allow administrators to register new Camp Managers and Field Staff accounts, capturing details such as full name, contact number, assigned region, and role. Supporters (Donors/Volunteers) shall be able to self-register.
- **Profile Management:** The system shall enable administrators to view, edit, and deactivate existing user profiles to maintain system security.

- **User Search & Filter:** The system shall provide search and filter functionalities for user records based on criteria such as role, location, or skill set (e.g., Medical Volunteer).
- **Order Management** (Refined as: Donation & Relief Tracking)
  - **Donation Pledge Creation:** The system shall allow Supporters to create new donation pledges, specifying the type (Cash/In-Kind), category (Food, Medicine, Shelter), quantity, and intended camp (optional).
  - **Relief Allocation:** The system shall allow Administrators to allocate pooled resources to specific Relief Camps based on reported needs.
  - **Status Tracking:** The system shall support tracking of donation status through predefined stages: "Pledged," "Received at HQ," "In Transit," and "Distributed."
  - **Donation Update:** The system shall allow authorized Camp Managers to update the status of received goods to acknowledge receipt.
  - **Search & Filter:** The system shall provide search and filter capabilities for donations based on donor ID, status, or item category.
- **Inventory/Materials Management**
  - **Resource Registration:** The system shall allow administrators to register standard relief items (e.g., Flour Bags, Tents, First Aid Kits) with details such as unit type and standard weight.
  - **Stock Tracking:** The system shall maintain real-time stock levels for all registered items within each specific Relief Camp.

- **Automatic Stock Decrement:** The system shall automatically decrease the camp stock level when resources are recorded as "Distributed" to an Affected Family by Field Staff.
- **Manual Stock Adjustment:** The system shall allow Camp Managers to manually adjust stock levels (e.g., for damaged goods or direct local donations).
- **Low-Stock Alerts:** The system shall generate automated alerts or notifications to the Admin when the stock level of critical items (e.g., clean water) falls below a minimum threshold at any camp.
- **Supplier/Source Details:** The system shall store information regarding the source of bulk relief supplies for audit purposes.
- **Employee/Labor Management**
  - **Volunteer Registration:** The system shall allow Supporters to register as Volunteers, capturing details such as skills (Doctor, Driver, General Labor) and availability.
  - **Task Assignment:** The system shall enable Camp Managers to assign specific tasks (e.g., "Unload Truck A," "Survey Sector B") to specific Volunteers or Field Staff.
  - **Task Status Tracking:** The system shall allow volunteers to update the status of their assigned tasks (e.g., "In Progress," "Completed").
  - **Performance Monitoring:** The system shall display the completion rate of tasks within a camp to identify bottlenecks in operations.
  - **Field Staff Assignment:** The system shall link Field Staff to specific geographic zones for victim data collection.

- **Billing & Payments**

- **Receipt Generation:** The system shall automatically generate digital acknowledgement receipts for Donors once their contribution is marked as "Received."
- **Distribution Recording:** The system shall allow Field Staff to record aid packages handed over to specific families, creating a digital audit trail.
- **Unmet Needs Tracking:** The system shall track the difference between "Requested Aid" by a camp and "Allocated Aid" by the Admin.
- **Resource Utilization Overview:** The system shall provide an overview of total resources received versus total resources distributed to verify zero-loss chains.

- **Reports & Analytics**

- **Dashboard:** The system shall provide a centralized dashboard displaying key metrics: Total Affected Families, Active Camps, and Urgent Resource Alerts.
- **Donation Reports:** The system shall generate reports on total contributions received by category (e.g., Total Liters of Water, Total Tents).
- **Camp Status Reports:** The system shall generate reports detailing current occupancy vs. capacity and current stock levels for each camp.
- **Distribution Reports:** The system shall provide reports on how many families have received aid vs. how many are still pending.
- **Volunteer Activity Reports:** The system shall generate reports on volunteer engagement and task completion rates.



- **Report Export:** The system shall allow reports to be exported in common formats (e.g., PDF, CSV) for government or NGO auditing.
- **User Authentication & Authorization**
  - **Secure Login:** The system shall provide a secure login mechanism for Admins, Managers, Staff, and Supporters.
  - **Role-Based Access Control (RBAC):** The system shall implement strict RBAC, ensuring:
    - Admins can see all data.
    - Camp Managers can only edit their specific camp data.
    - Field Staff can only access victim data entry forms.
    - Supporters can only view their own history and public appeals.
  - **Password Management:** The system shall allow users to securely reset their forgotten passwords.

## **NON-FUNCTIONAL REQUIREMENTS: -**

- **Performance**
  - **Response Time:** The system shall respond to user requests (e.g., loading the donation dashboard) within 3 seconds under normal load conditions.
  - **Data Processing Speed:** Reports involving camp inventory aggregation shall be generated within 10 seconds.
  - **Concurrency:** The system shall support at least 50 concurrent users (Admins, Managers, Staff) without degradation, as relief operations can be intense.

- **Security**

- **Data Encryption:** Sensitive data (e.g., passwords, victim CNIC numbers) shall be hashed or encrypted in the database.
- **Access Control:** The system shall rigorously enforce permissions so that Donors cannot see private data of Affected Families.
- **Input Validation:** The system shall perform robust input validation to prevent SQL injection and XSS attacks, ensuring the database remains secure.
- **Audit Trails:** The system shall maintain an audit trail for critical actions (e.g., stock adjustments, deletion of family records).

- **Usability**

- **Intuitive User Interface:** The UI (built with Bootstrap) shall be clean and simple, enabling Field Staff with limited technical training to use it easily.
- **Mobile Responsiveness:** The system must be fully responsive to allow Field Staff to access it via smartphones/tablets on-site.
- **Error Handling:** The system shall provide clear, non-technical error messages (e.g., "Registration Failed: Email already in use").
- **Learnability:** A new Camp Manager should be able to perform basic tasks (e.g., update camp capacity) within 30 minutes of training.

- **Reliability**

- **System Uptime:** The system shall aim for high availability (99.5%) to ensure it is accessible during critical disaster response windows.
- **Data Integrity:** The system shall ensure that inventory numbers are accurate and consistent (e.g., stock cannot be negative).
- **Backup & Recovery:** A mechanism for database export/backup (MySQL Dump) shall be available to prevent data loss.

- **Scalability**

- **User Growth:** The system shall be designed to accommodate a sudden influx of volunteer registrations during a disaster event without crashing.
- **Data Storage:** The MySQL database architecture shall be capable of handling thousands of family records and donation logs efficiently.

- **Maintainability**

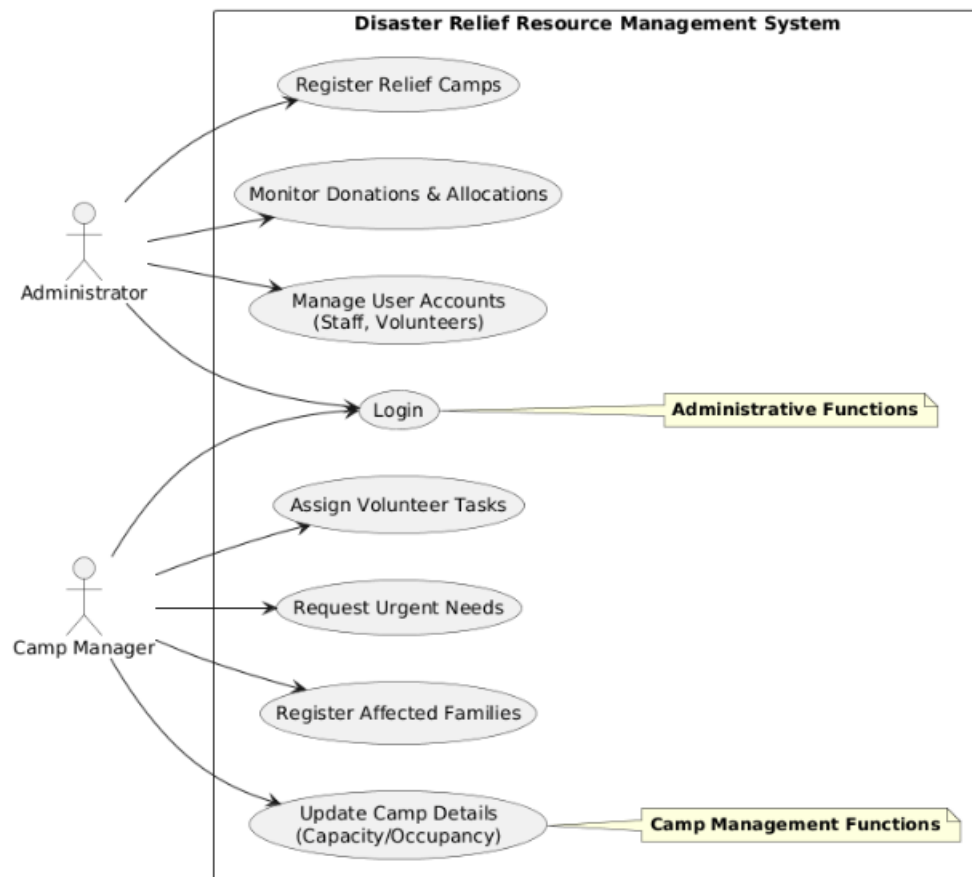
- **Code Quality:** The PHP code shall be modular (separating logic for Admin, Manager, and Staff) to facilitate future updates.
- **Documentation:** Comprehensive documentation (SRS, User Manual) shall be provided.
- **Logging:** The system shall log PHP errors to assist developers in debugging issues during deployment.

- **Compatibility**

- **Browser Compatibility:** The web application shall be fully functional on modern web browsers (Chrome, Firefox, Edge, Safari) and mobile browsers.

- **Operating System Compatibility:** The system shall operate independently of the client's OS (Windows, Android, macOS, Linux).

### Use Case Diagram(s):



**Usage Scenarios: -**

**Use Case: Register New Relief Camp**

| <b>Field</b>             | <b>Details</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Abbreviated Title</b> | Reg. New Camp                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Use Case ID</b>       | UCM-001                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Description</b>       | Enables an administrator to register a new relief camp location into the system, establishing a centralized profile for tracking inventory, occupancy, and volunteer assignments.                                                                                                                                                                                                                                                                                     |
| <b>Pre-Conditions</b>    | Administrator is logged into the system and has navigated to the Camp Management module.                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Task Sequence</b>     | <ol style="list-style-type: none"><li>1. Administrator clicks on the "Relief Camps" tab/menu item.</li><li>2. Administrator selects "Add New Camp" option.</li><li>3. Administrator enters required camp details (e.g., Camp Name, District, GPS Coordinates, Capacity, Assigned Manager).</li><li>4. Administrator reviews entered details and clicks "Save" or "Register".</li><li>5. System displays a success message confirming the camp registration.</li></ol> |
| <b>Exceptions</b>        | <ul style="list-style-type: none"><li>- Required fields (e.g., Location, Capacity) are left blank.</li><li>- Invalid data format (e.g., negative capacity number).</li></ul>                                                                                                                                                                                                                                                                                          |

|                             |                                                                                           |
|-----------------------------|-------------------------------------------------------------------------------------------|
|                             | - A camp with an identical name or location ID already exists.                            |
| <b>Post Conditions</b>      | A new relief camp profile is created in the database, accessible for resource allocation. |
| <b>Unresolved Issues</b>    | N/A                                                                                       |
| <b>Authority</b>            | Administrator                                                                             |
| <b>Modification History</b> | 1.0 (Initial Draft)                                                                       |
| <b>Author</b>               | -                                                                                         |

#### **Use Case: Record Urgent Needs**

|                          |                                                                                                                                        |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Field</b>             | <b>Details</b>                                                                                                                         |
| <b>Abbreviated Title</b> | Req. Urgent Needs                                                                                                                      |
| <b>Use Case ID</b>       | UCM-002                                                                                                                                |
| <b>Description</b>       | Enables a Camp Manager to post an urgent appeal for specific supplies (e.g., food, medicine) required at their specific camp location. |
| <b>Pre-Conditions</b>    | Camp Manager is logged into the system and is assigned to an active relief camp.                                                       |
| <b>Task Sequence</b>     | 1. Camp Manager clicks on the "Inventory & Needs" module.                                                                              |

|                             |                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | <p>2. Manager selects "Create Urgent Request".</p> <p>3. Manager selects the item category (e.g., Medicine), item name, and required quantity.</p> <p>4. Manager sets the priority level (High/Critical) and adds a description.</p> <p>5. Manager clicks "Submit Request".</p> <p>6. System updates the dashboard and alerts the Administrator.</p> |
| <b>Exceptions</b>           | <ul style="list-style-type: none"> <li>- Quantity entered is zero or invalid.</li> <li>- Item selected is not in the master database.</li> <li>- Internet connection failure prevents submission.</li> </ul>                                                                                                                                         |
| <b>Post Conditions</b>      | An alert is generated on the Admin Dashboard and the request is logged in the "Pending Allocations" list.                                                                                                                                                                                                                                            |
| <b>Unresolved Issues</b>    | N/A                                                                                                                                                                                                                                                                                                                                                  |
| <b>Authority</b>            | Camp Manager                                                                                                                                                                                                                                                                                                                                         |
| <b>Modification History</b> | 1.0 (Initial Draft)                                                                                                                                                                                                                                                                                                                                  |
| <b>Author</b>               | -                                                                                                                                                                                                                                                                                                                                                    |



### Use Case: Submit Donation

| Field             | Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Abbreviated Title | Submit Donation                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Use Case ID       | UCM-003                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Description       | Enables a Supporter (Donor) to pledge a donation (Cash or In-Kind) and log it into the system for tracking.                                                                                                                                                                                                                                                                                                                                                                               |
| Pre-Conditions    | Supporter is registered and logged into the web application.                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Task Sequence     | <ol style="list-style-type: none"><li>1. Supporter navigates to the "Donate Now" section.</li><li>2. Supporter selects donation type (Cash / Goods).</li><li>3. If Goods: Supporter enters item details (e.g., Blankets, Rice), quantity, and drop-off preference.</li><li>4. If Cash: Supporter enters amount and payment reference details.</li><li>5. Supporter clicks "Pledge Donation".</li><li>6. System generates a unique Donation ID and status as "Pledged/Received".</li></ol> |
| Exceptions        | <ul style="list-style-type: none"><li>- Invalid quantity or amount entered.</li><li>- Missing mandatory fields (e.g., item description).</li></ul>                                                                                                                                                                                                                                                                                                                                        |
| Post Conditions   | The donation is recorded in the database and visible to Administrators for allocation to camps.                                                                                                                                                                                                                                                                                                                                                                                           |
| Unresolved Issues | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|                             |                     |
|-----------------------------|---------------------|
| <b>Authority</b>            | Supporter (Donor)   |
| <b>Modification History</b> | 1.0 (Initial Draft) |
| <b>Author</b>               | -                   |

### **Use Case: Register Affected Family**

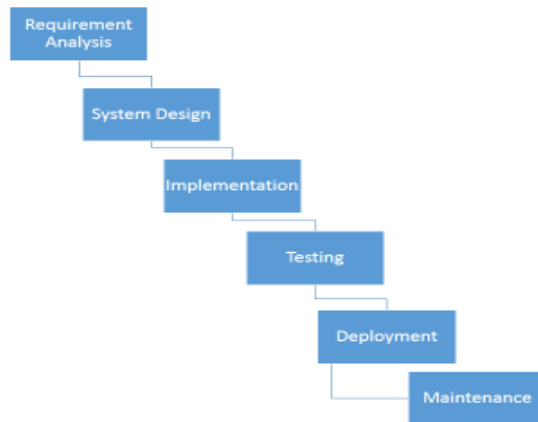
|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Field</b>             | <b>Details</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Abbreviated Title</b> | Reg. Family                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Use Case ID</b>       | UCM-004                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Description</b>       | Enables Field Staff to register a victim family unit, recording their personal details, losses, and immediate needs for aid distribution.                                                                                                                                                                                                                                                                                                                                        |
| <b>Pre-Conditions</b>    | Field Staff is logged into the system and physically present at the assessment site/camp.                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Task Sequence</b>     | <ol style="list-style-type: none"> <li>1. Field Staff selects "Family Registration" from the mobile-responsive menu.</li> <li>2. Staff enters Head of Household details (Name, CNIC, Phone).</li> <li>3. Staff records family size and specific damages (e.g., House Destroyed: Yes).</li> <li>4. Staff tags any special medical needs (e.g., Elderly, Injured).</li> <li>5. Staff clicks "Register Family".</li> <li>6. System saves the record and assigns a unique</li> </ol> |

|                             |                                                                                                                                                                                                                                            |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | Family ID.                                                                                                                                                                                                                                 |
| <b>Exceptions</b>           | <ul style="list-style-type: none"> <li>- CNIC entered is already registered in the system (Duplicate entry).</li> <li>- Mandatory fields (Family Size, Name) are missing.</li> <li>- Database timeout due to poor connectivity.</li> </ul> |
| <b>Post Conditions</b>      | The family is added to the beneficiary database and becomes eligible for aid distribution tracking.                                                                                                                                        |
| <b>Unresolved Issues</b>    | N/A                                                                                                                                                                                                                                        |
| <b>Authority</b>            | Field Staff                                                                                                                                                                                                                                |
| <b>Modification History</b> | 1.0 (Initial Draft)                                                                                                                                                                                                                        |
| <b>Author</b>               | -                                                                                                                                                                                                                                          |

## Adopted Methodology

In the modern era of software engineering, selecting an appropriate development methodology is critical to the success of a project. While there are numerous methodologies available, we have selected a hybrid approach for the **Disaster Relief Resource Management System**. This approach is known as the **VU Process Model**, which combines the structural discipline of the **Waterfall Model** with the iterative nature of the **Spiral Model**.

### Waterfall Model



## EXPLANATION OF WATERFALL AND SPIRAL METHODOLOGY

### WATERFALL MODEL:-

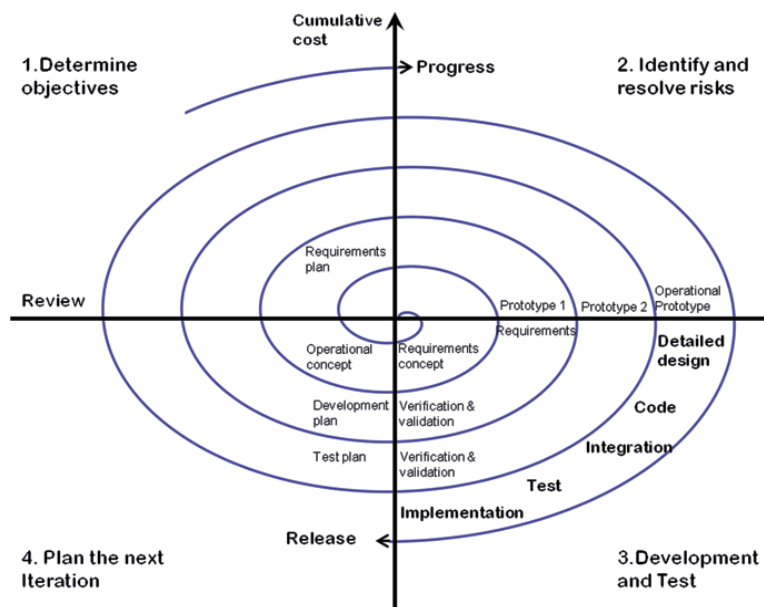
The Waterfall model is a classical software development methodology used in the System Development Life Cycle (SDLC) to create a system using a linear and sequential approach. In this model, the development process flows steadily downwards (like a waterfall) through several phases.

- **System Services & Goals:** Requirements, such as the need for Admin and Camp Manager modules, are well-defined at the start.

- **Planning:** It provides a complete blueprint of the system before coding begins.
- **Implementation:** The system is developed, installed, and put into practical use.
- **Maintenance:** The system is checked for errors, corrected, and improved after deployment.

### **SPIRAL MODEL:-**

This methodology combines elements of the Waterfall model with the iterative nature of prototyping. It places a heavy emphasis on **Risk Analysis**. A software project repeatedly passes through phases in iterations (spirals). The core phases are: Planning, Risk Analysis, Engineering (Constructing), and Evaluation. This is particularly useful for complex projects where requirements might evolve.



### **ADOPTED MODEL / METHODOLOGY AS VU SPIRAL MODEL:-**

#### **REASON:-**

For the development of the **Disaster Relief Resource Management System**, we have adopted the **VU Process Model**. This is a **Hybrid Approach** that leverages the strengths of both Waterfall and Spiral methodologies. It ensures that the project follows a logical sequence while allowing for feedback and refinement.

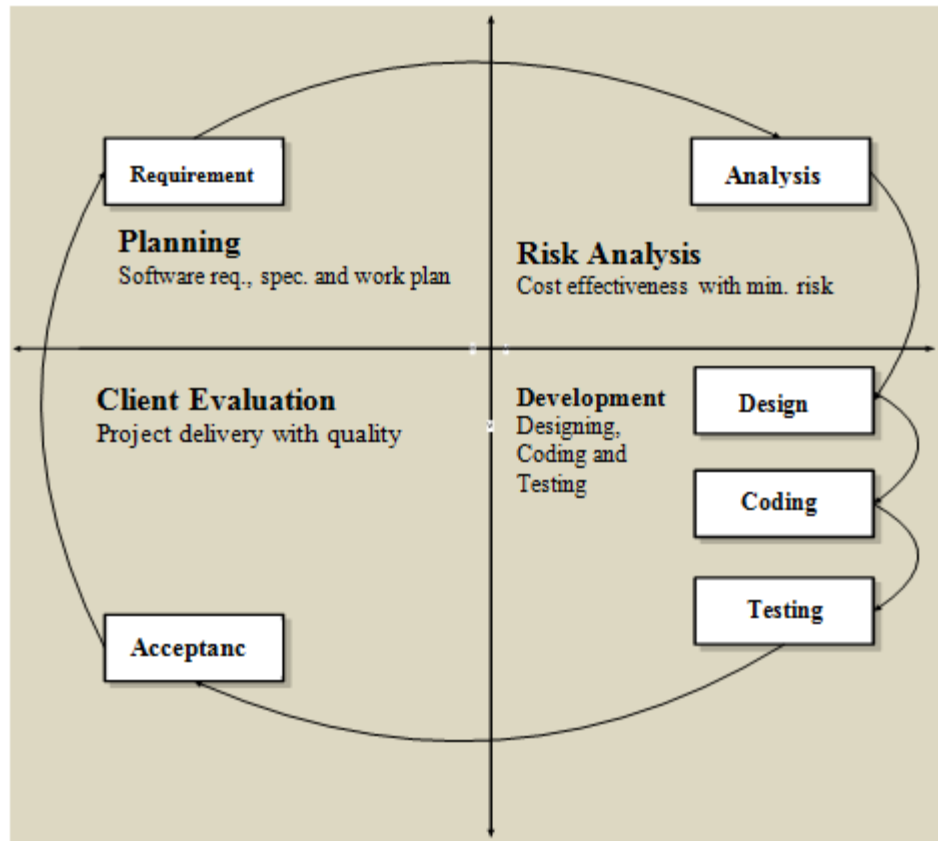
The project lifecycle is divided into the following five (05) specific phases:

1. **Requirements Gathering & Analyzing** (Creating the SRS for relief operations).
2. **Planning** (Scheduling tasks and defining scope).
3. **Analysis and Design** (Creating database schemas, Use Cases, and UI designs).
4. **Development** (Coding the PHP/MySQL modules).
5. **Final Report & Delivery** (Testing and deployment).

#### **How the Hybrid Model Works:**

The project is split into the multiple phases mentioned above. We complete every phase in a sequence (Waterfall approach) and submit the deliverables to the **concerned supervisor**. If the supervisor suggests improvements or identifies risks in any phase (e.g., changes to the Donation Tracking logic), we revisit and refine that specific phase (Spiral approach).

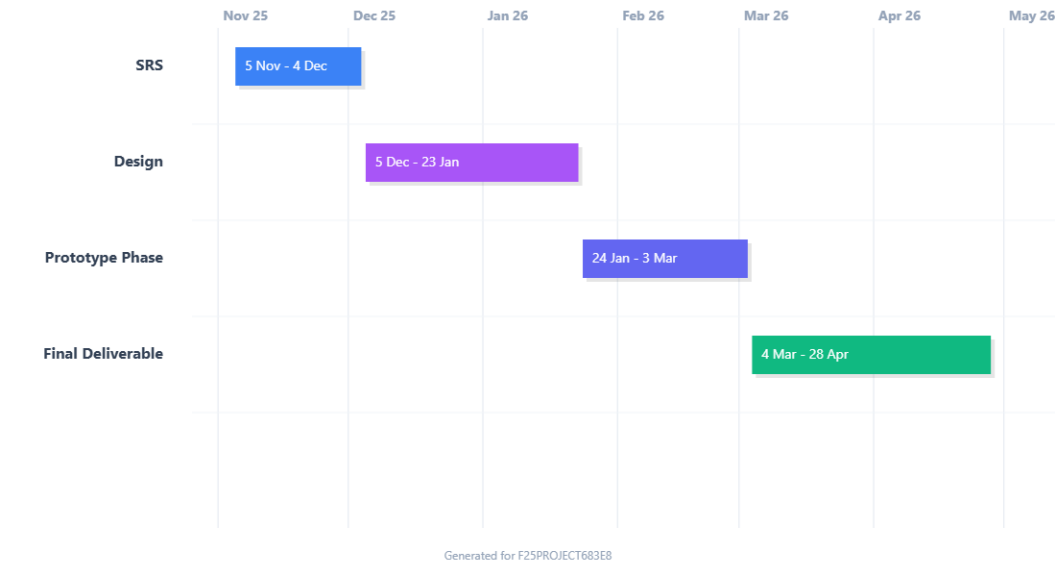
Once the current phase is approved according to the instructions of the Supervisor, the next phase is processed. This ensures that the **Disaster Relief Resource Management System** is built on a solid foundation where errors and bugs are minimized, as every step is verified and completed in sequence before moving forward.



**Work Plan**

**Disaster Relief Resource Management System**

Group ID: F25PROJECT683E8



\*\*\*\*\* **End of the Document** \*\*\*\*\*